

CARANSEBES, Romania

WMO: 152920

Lat: 45.420N

Lon: 22.253E

Elev: 264

StdP: 98.19

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-11.0	-8.4	-13.9	1.2	-9.3	-11.3	1.5	-7.0	11.7	1.4	10.1	2.0	1.2	150	0.464

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	12.6	33.1	21.0	31.4	21.0	29.7	20.4	23.1	29.9	22.1	28.9	21.1	27.8	2.8	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.8	16.0	26.8	19.8	15.0	25.0	19.0	14.2	23.9	69.4	29.8	65.6	28.9	62.3	27.8	28.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.5	7.5	6.4	DB	-15.4	35.2	3.5	2.2	-18.0	36.7	-20.0	38.0	-22.0	39.2	-24.5	40.8	(4)
(5)			WB	-15.7	24.6	3.4	1.4	-18.1	25.6	-20.1	26.5	-22.0	27.2	-24.5	28.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	11.5	0.4	2.4	6.5	12.1	16.3	19.5	21.4	21.6	16.8	12.0	7.3	1.7	(6)	
	DBStd	8.52	4.87	5.04	4.59	3.99	3.59	3.43	3.17	3.28	3.84	4.37	4.94	4.79	(7)	
	HDD10.0	1062	297	214	127	24	2	0	0	0	1	31	109	258	(8)	
	HDD18.3	2803	554	446	367	190	83	27	9	8	73	199	331	515	(9)	
	CDD10.0	1627	1	2	18	86	196	286	353	359	205	92	28	2	(10)	
	CDD18.3	328	0	0	0	2	19	63	103	109	27	3	0	0	(11)	
	CDH23.3	3512	0	0	1	36	214	681	1110	1163	274	32	1	0	(12)	
	CDH26.7	1276	0	0	0	4	35	222	445	489	77	3	0	0	(13)	
(14) Wind	WSAvg	2.1	2.2	2.5	2.5	2.5	2.0	1.8	1.7	1.8	1.9	2.2	2.7	2.1	(14)	
(15) Precipitation	PrecAvg	882	48	52	52	73	85	115	107	93	88	62	57	57	(15)	
	PrecMax	1301	88	165	96	188	193	203	192	257	210	134	149	127	(16)	
	PrecMin	613	16	8	11	10	30	24	38	15	19	4	13	16	(17)	
	PrecStd	184	20	36	23	40	39	47	44	62	45	36	33	30	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.8	17.1	21.2	26.8	29.5	33.0	35.6	36.0	32.1	26.2	21.1	14.6	(19)	
		MCWB	8.8	10.4	11.8	15.9	19.1	21.6	20.6	21.2	19.0	16.7	13.2	9.6	(20)	
	2%	DB	11.0	13.6	18.5	24.0	27.5	31.1	33.0	33.6	28.8	23.8	18.2	12.1	(21)	
		MCWB	7.5	8.6	10.7	14.9	18.2	21.4	21.0	21.0	18.8	15.8	12.5	8.4	(22)	
	5%	DB	8.8	11.3	16.1	21.5	25.8	29.2	31.1	31.4	26.4	21.5	16.1	10.1	(23)	
		MCWB	6.3	7.3	9.6	13.2	17.4	20.7	21.1	20.8	17.8	15.0	11.2	7.3	(24)	
	10%	DB	6.8	9.3	13.9	19.4	24.0	27.4	29.1	29.4	24.2	19.1	14.2	8.2	(25)	
		MCWB	5.1	6.3	8.6	12.2	16.7	19.8	20.6	20.2	17.2	14.0	10.3	6.0	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.4	10.8	12.9	17.0	21.0	23.9	24.1	24.5	21.5	18.3	14.2	10.4	(27)	
		MCDB	12.9	15.6	19.5	25.3	27.4	30.5	30.4	31.4	28.8	23.5	19.0	12.8	(28)	
	2%	WB	7.9	9.1	11.4	15.5	19.3	22.3	23.0	22.9	19.7	16.7	13.1	8.8	(29)	
		MCDB	10.4	12.8	16.8	22.5	25.3	29.2	29.5	30.4	25.5	22.0	17.1	11.7	(30)	
	5%	WB	6.5	7.8	10.2	14.1	18.2	21.2	22.0	21.7	18.7	15.6	12.0	7.5	(31)	
		MCDB	8.5	10.9	15.1	20.1	24.1	27.8	28.7	28.6	24.3	20.3	15.4	9.8	(32)	
	10%	WB	5.1	6.5	9.1	12.9	17.2	20.2	21.1	20.7	17.6	14.4	10.8	6.2	(33)	
		MCDB	6.7	9.0	13.2	17.9	22.8	26.4	27.6	27.4	23.0	18.5	13.9	8.1	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	6.4	7.6	9.6	10.7	11.3	11.7	12.3	12.6	10.9	10.2	7.8	6.1	(35)	
		MCDBR	9.5	10.7	12.8	14.1	14.3	14.6	15.5	15.8	14.6	13.5	10.3	9.0	(36)	
	5% WB	MCWBR	6.1	6.6	7.0	6.8	7.0	6.8	6.6	6.3	6.3	6.8	5.8	5.9	(37)	
		MCWBR	8.7	9.8	11.5	12.7	13.2	13.5	13.6	13.5	12.5	11.9	9.6	8.3	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.329	0.359	0.403	0.463	0.449	0.439	0.454	0.468	0.431	0.388	0.355	0.329	(40)	
		taud	2.304	2.214	2.172	2.035	2.139	2.214	2.175	2.129	2.198	2.310	2.348	2.342	(41)	
	Ebn at Noon		760	801	817	796	824	833	814	783	779	762	723	719	(42)	
		Edh at Noon	81	107	128	160	149	139	143	144	124	97	78	71	(43)	
(44) All-Sky Solar Radiation	RadAvg		1.29	2.09	3.24	4.41	5.27	5.91	5.86	5.29	3.75	2.60	1.54	1.06	(44)	
	RadStd		0.16	0.27	0.44	0.48	0.48	0.56	0.47	0.52	0.39	0.30	0.18	0.14	(45)	

Historical Trends

		Heating		Cooling			Degree-Days					
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	+0.72	N/A	N/A	+1.13	N/A	N/A	N/A	-160	+96	N/A	(46)
(47)	Regional (71 neighbors)	+0.68	N/A	N/A	N/A	N/A	N/A	-117	-180	+121	+62	(47)

Nomenclature: See separate page